



# Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

|              |    |              |      |
|--------------|----|--------------|------|
| Class:       | BE | Semester:    | VIII |
| Course Code: |    | Course Name: | RL   |

|                      |                  |
|----------------------|------------------|
| Name of Student:     | BARI ANKIT VINOD |
| Roll No. :           | 61               |
| Assignment No.:      | 4                |
| Title of Assignment: |                  |
| Date of Submission:  |                  |
| Date of Correction:  |                  |

## Evaluation

| Performance Indicator  | Max. Marks | Marks Obtained |
|------------------------|------------|----------------|
| Completeness           | 5          | 04             |
| Demonstrated Knowledge | 3          | 02             |
| Legibility             | 2          | 02             |
| Total                  | 10         | 08             |

| Performance Indicator             | Exceed Expectations (EE) | Meet Expectations (ME) | Below Expectations (BE) |
|-----------------------------------|--------------------------|------------------------|-------------------------|
| Completeness                      | 5                        | 3-4                    | 1-2                     |
| Demonstrated Knowledge Legibility | 3                        | 2                      | 1                       |
| Legibility                        | 2                        | 1                      | 0                       |

Checked by

Name of Faculty : Mrs. Sneha Yadav

Signature :

Date :

1) Apply generalized policy iteration in reinforcement learning environment and show policy evaluation and policy improvement work together to improve the agent's performance.

→ Generalized policy iteration (GPI) is the interaction between -

- policy evaluation - estimate how good the current policy is
- policy improvement - improve the policy using those estimates.

- Eg. grid world navigation -

- Environment -  $4 \times 4$  grid

- start at top-left, goal at bottom-right

- reward - -1 per step

- 0 at goal

31- Initialize policy - start with random policy ( $\pi_0$ )

32- Policy evaluation -  $V(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a) [R + \gamma V(s')]$

- Iterate until values stabilized

- eg. values after evaluation.

33- Policy improvement -  $\pi'(s) = \underset{a}{\operatorname{argmax}} \sum_{s'} P(s'|s,a) [R + \gamma V(s')]$

- choose action that leads to higher value state.

34- Repeat (GPI loop) - new policy → evaluate again

- improved values → improve policy again

- final outcome - policy convergence to optimal policy ( $\pi^*$ )

- Agent learns shortest path to goal.

Q. 2) Demonstrate the use of asynchronous dynamic programming in a robot navigation problem where only frequently visited states are updated.

→ Asynchronous DP - updates only selected states  
- often those frequently visited.

- Eg. Robot navigation -

- Environment - warehouse grid

- Robot frequently visits

- charging station

- loading zones

- Rarely visits - corners or blocked paths.

- Idea - Instead of updating all states.

- update only states in robot's actual trajectory.

- Algorithm (Asynchronous value iteration)

$$V(s) \leftarrow \max_a \sum_{s'} p(s'/s, a) [R + \gamma V(s')]$$

31. Initialize values - All states  $V(s) = 0$

32. Robot runs - start  $\rightarrow A \rightarrow B \rightarrow$  charging  $\rightarrow A \rightarrow$  Goal

33. Update only visited states - Instead of full sweep update  $V(\text{start}), V(A), V(B), V(\text{charging})$

34. Repeated over time - As robot explores -

- frequently visited states get many updates

- Rare states get few updates.